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DSC 140 Final Project

Strokes are the fifth leading cause of death in the U.S. and a leading cause of long-term disability world wide. Identifying risk factors is important to understand who may be at a higher chance to a stroke. Then they can take preventative measures according to their doctor. The purpose of this project is to explore a Stroke dataset and be able to answer meaningful questions based on the relationships between medical and lifestyle factors on the occurrence of a stroke. This dataset was acquired from Kaggle. It contains 5,110 rows of patient records with various types of data including but not limited to: age, BMI, glucose level, gender, work type, and hypertension. With the mix of continuous, binary, and categorical variables it is suitable to analyze and answer questions. In this project the following research questions I explored include: what variables best predict a stroke, can you identify different health profiles using clustering, and does average glucose level differ between patients who had strokes vs who haven't across professions.

The first step to the project is data cleaning and preprocessing. This consist of removing missing values as they can cause error when computing calculations and doing machine learning. I decided to start with the dataset in RStudio as preprocessing is simpler for me compared to Python. When examining the updated dataset, I expected rows to be removed due to "N/A" values in the BMI column. However, no rows were dropped because "N/A" is stored as a string, not as an actual missing value. What I did was remove all rows with the value "N/A". Next, I removed all rows with the value "Unknown" as that was a percentage of the smoking status column. Lastly, I removed all rows with "Other" in the gender column as that does not help me process information and learn from the dataset. With all values that would interfere with the interpretation of the dataset I could move onto creating new columns to further improve the interpretability. There were already three binary variables of stroke, hypertension, and heart disease. I created corresponding YES/NO columns to make these variables easier to understand. My last step to complete the preprocessing of the dataset was to insure all columns were the correct data type. As our BMI column which contained numbers also contained a string of "N/A", that column was not numerical as it should've been. With all the string values removed I could now convert the BMI column to numeric which allows me to analyze it as actual numbers. Below is a picture of the processed dataset:

	id	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	stroke	stroke_cat	hypertension_cat	heart_disease_cat
1	9046	Male	67	0	1	Yes	Private	Urban	228.69	36.6	formerly smoked	1	Yes	No	Yes
2	31112	Male	80	0	1	Yes	Private	Rural	105.92	32.5	never smoked	1	Yes	No	Yes
3	60182	Female	49	0	0	Yes	Private	Urban	171.23	34.4	smokes	1	Yes	No	No
4	1665	Female	79	1	0	Yes	Self-employed	Rural	174.12	24.0	never smoked	1	Yes	Yes	No
5	56669	Male	81	0	0	Yes	Private	Urban	186.21	29.0	formerly smoked	1	Yes	No	No
6	53882	Male	74	1	1	Yes	Private	Rural	70.09	27.4	never smoked	1	Yes	Yes	Yes
7	10434	Female	69	0	0	No	Private	Urban	94.39	22.8	never smoked	1	Yes	No	No
8	12109	Female	81	1	0	Yes	Private	Rural	80.43	29.7	never smoked	1	Yes	Yes	No
9	12095	Female	61	0	1	Yes	Govt_job	Rural	120.46	36.8	smokes	1	Yes	No	Yes
10	12175	Female	54	0	0	Yes	Private	Urban	104.51	27.3	smokes	1	Yes	No	No
11	5317	Female	79	0	1	Yes	Private	Urban	214.09	28.2	never smoked	1	Yes	No	Yes
12	58202	Female	50	1	0	Yes	Self-employed	Rural	167.41	30.9	never smoked	1	Yes	Yes	No
13	56112	Male	64	0	1	Yes	Private	Urban	191.61	37.5	smokes	1	Yes	No	Yes
14	34120	Male	75	1	0	Yes	Private	Urban	221.29	25.8	smokes	1	Yes	Yes	No
15	27458	Female	60	0	0	No	Private	Urban	89.22	37.8	never smoked	1	Yes	No	No
16	70630	Female	71	0	0	Yes	Govt_job	Rural	193.94	22.4	smokes	1	Yes	No	No
17	13861	Female	52	1	0	Yes	Self-employed	Urban	233.29	48.9	never smoked	1	Yes	Yes	No
18	68794	Female	79	0	0	Yes	Self-employed	Urban	228.70	26.6	never smoked	1	Yes	No	No
19	4219	Male	71	0	0	Yes	Private	Urban	102.87	27.2	formerly smoked	1	Yes	No	No

Figure 1: Processed Stroke Dataset

The processed dataset contains 15 columns and 3,425 rows. The number of rows and columns is sufficient enough to be able to answer a multitude of questions. Before I can answer questions, I need to first find out information about the variables specifically and the relationships between them. I first wanted to see how age varied throughout the dataset. I created a histogram of the age distribution which showed a wide range of patients between ten years old to eighty-two years old. The histogram is shown below:

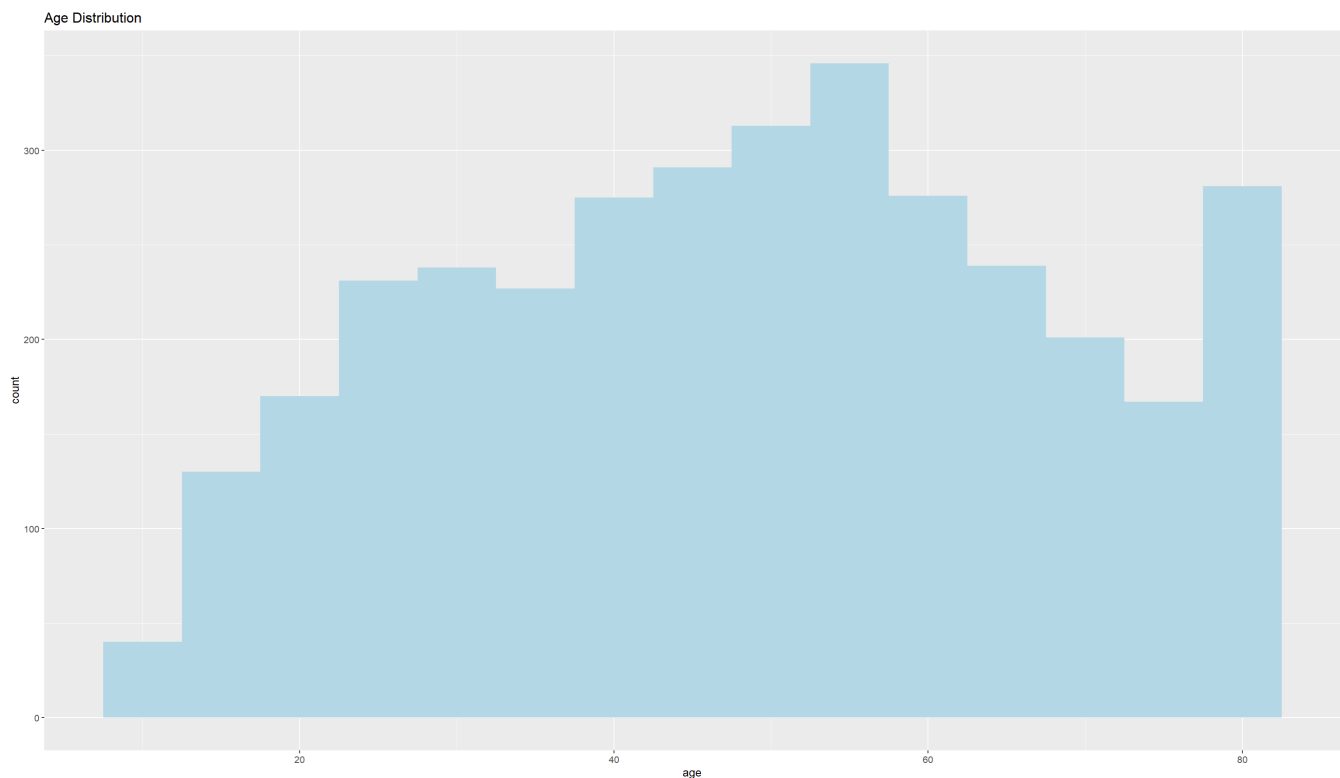


Figure 2: Age distribution across the dataset

With this we are able to see the majority of the patients are between the ages of 40 to 60 years old. The majority of the patients being between 52 and 58 years old. The wide variety of patients allows us to examine the outcome across different age groups and ensure that our analysis isn't biased towards a certain age group.

Next, I wanted to find out what variables correlated with each other to help determine a machine learning algorithm and questions that could be asked. To receive this information I performed a chi-squared-test. A chi-squared-test is a statistical test which determines whether two categorical variables are associated with each other. When completing the test the important value to look for is the p-value. The p-value tells you how likely it is that those values correlate with each other. If the p-value is less than 0.05 that means the values are associated with each other and if it greater than 0.05 then there is no association. In this dataset I always compared the variables to stroke as that is my main focus. I completed the chi-squared-test three separate times comparing profession, smoking status, hypertension all to stroke. The p-value for profession and stroke is 0.0128. The p-value for smoking status and stroke is 0.04906. The p-value for hypertension and stroke is $2.2e-16$. The p-value for heart disease and stroke is $1.1915e-15$. With this information we know that smoking status, hypertension, and heart disease all correlate with stroke. This means that patients who smoke are more likely to have a stroke compared to those who don't. Patients who have hypertension and heart disease are also more likely to have a stroke compared to patients without hypertension and heart disease. I also solved for pearsons correlation coefficient. This measures the strength between two numerical variables. It ranges from "-1" to "1" where "1" is a perfect positive linear relationship, "-1" is a perfect negative linear relationship, and "0" means there is no linear relationship. The correlation between age and average glucose level outputted a value of 0.234 which means there is a weak positive correlation. This means that older individuals tend to have slightly higher average glucose levels, but age alone is not a strong predictor of glucose. Now that we analyzed the initial part of the dataset I can create a machine learning algorithm. I chose to create a machine learning algorithm in Python as it is more convenient and easier to use.

The machine learning algorithm I chose to create was a neural network. A neural network is a computer model that learns patterns from data by taking selected inputs and processing them through hidden layers made up of interconnected neurons. Each layer learns a piece of the pattern, and when they are combined, create a final prediction. I used the neural network to predict whether a patient would have a stroke. The inputs for my neural network included age, average glucose level, BMI, hypertension, and heart disease. All the numerical features were scaled and the binary numbers were left alone. The output was a binary prediction of stroke, "1", and no stroke, "0". I used hyperparameter tuning which is a process of testing different values of the number of hidden layers and neurons, activation function, and the number of iterations to see which combination produces the best results. The neural network performed best using a logistic activation function, 100 training iterations, and a hidden layer architecture of three layers with 5 neurons each.

This produced results of 95% best accuracy and 93% test accuracy. To better showcase these results a confusion matrix was created. A confusion matrix is a table that shows how well the classification model made its predictions by comparing actual to predicted.

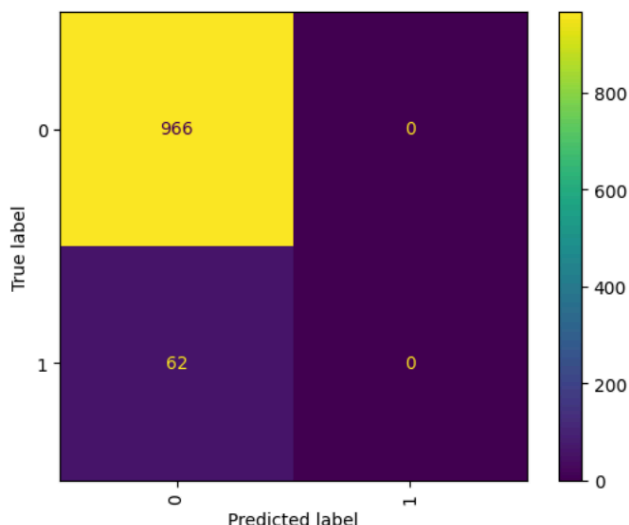


Figure 3: Confusion matrix of neural network

Analyzing the confusion matrix, we can see that the model predicted "no stroke" 1028 times out of 1028 predictions. With predicted 966 of the 1028 correctly it leads to our percentage of 93%. That accuracy is amazing, however there is a problem with this neural network. It never once predicted "stroke". With this I can assume the neural net failed due to a multitude of reasons. First, the dataset is imbalanced. In the dataset the majority of the the stroke column is "1", specifically 95% of the dataset. So the neural network can predict "0" every time and still be correct roughly 95% of the time. Also, the wrong evaluation metric may have been used as, accuracy, makes the model optimize trying to achieve the best percentages. The next step in the project was to answer questions.

The first question I answered, what are the most important variables when predicting a stroke. I chose this question because it is relevant when understanding what factor influence stroke risk. If we are able to identify these variables then we can help prioritize helping people lower their risk factor through lifestyle and health changes. To answer this question, I used a new method of analysis called random forest algorithm. Random forest algorithm is a machine learning model used for classification or regression which uses many decision trees to make better predictions. Each decision tree is slightly different from each other and gives its own prediction based on its random part of the data. For classification it chooses the final answer based on what the majority of the decision trees predicted. For regression it chooses the final number as the average of all the trees predictions. I chose to use the random forest algorithm for classification as I was predicting stroke.

The variables I chose to include were based on information found out when initially learning about the dataset. The variables I included were average glucose level, BMI, hypertension, heart disease, and if they were ever married. I created a bar graph to better show the results found from the machine learning model.

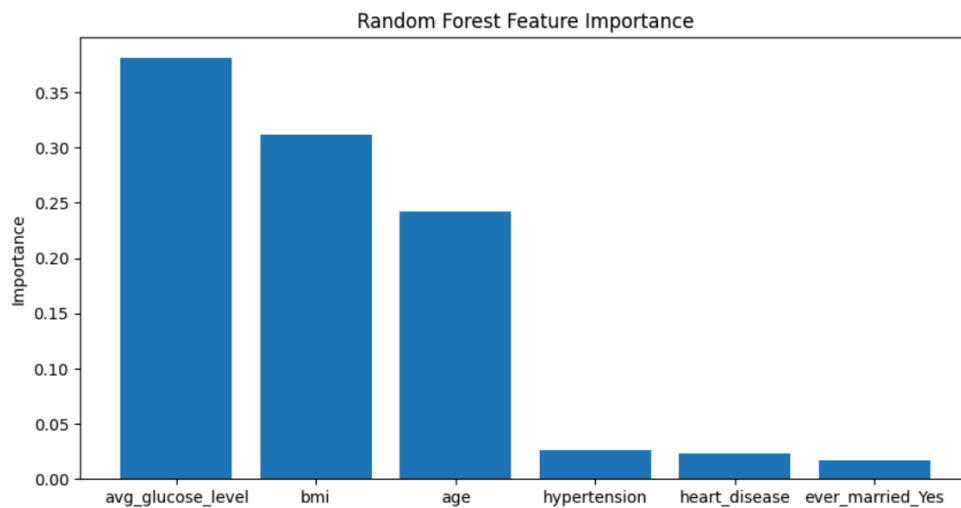


Figure 4: Feature importance when predicting stroke

From the graph above we can see that three variables stand out the most for importance when predicting for a stroke. These three variables include average glucose level, BMI, and age. Glucose level had an importance of 38.1%, BMI had an importance of 31.2%, and age had an importance of 24.3%. The glucose level being 8.1% means it is the most influential factor when predicting stroke in this dataset. A high glucose level can be a very strong indication of a patient having diabetes. Diabetes is a condition that can increase the risk for stroke as it can cause sugars to build up in the bloodstream and prevent oxygen from reaching the brain. Another important feature is BMI, patients with a high BMI, specifically above 30 can be considered obese. Obesity also increases the risk for stroke as it can lead to diabetes and blood pressure. Hypertension, also known as high blood pressure had an importance of 2.6%. Based on the dataset hypertension is not a important factor when predicting for stroke. However, based on scientific information high blood pressure is critically important as it significantly increases risk for serious conditions such as strokes. Now that we have identified the most important features when predicating a stroke, we can further explore how these factors cluster patients into different health profiles.

To group patients together another new method was used called hierarchical clustering, which groups individual based on similarities across key features. Hierarchical clustering is an unsupervised machine learning model which builds a dendrogram that shows how clusters merge or split at different levels based on the similarity across the selected features. Hierarchical clustering starts by assigning each patient as its own cluster, then merges the most similar clusters until all the patients are combined into a certain number of

clusters requested. I needed to determine the optimal number of clusters to be able to interpret the dendrogram. I applied KMeans clustering and visualized it on an elbow plot to determine the optimal number of clusters. These clusters were created based on the inputs variables of age, glucose level, BMI, and hypertension. The analysis identified four clusters which means that the dataset naturally groups into 4 subgroups based on the data. With this information we can properly create the dendrogram.

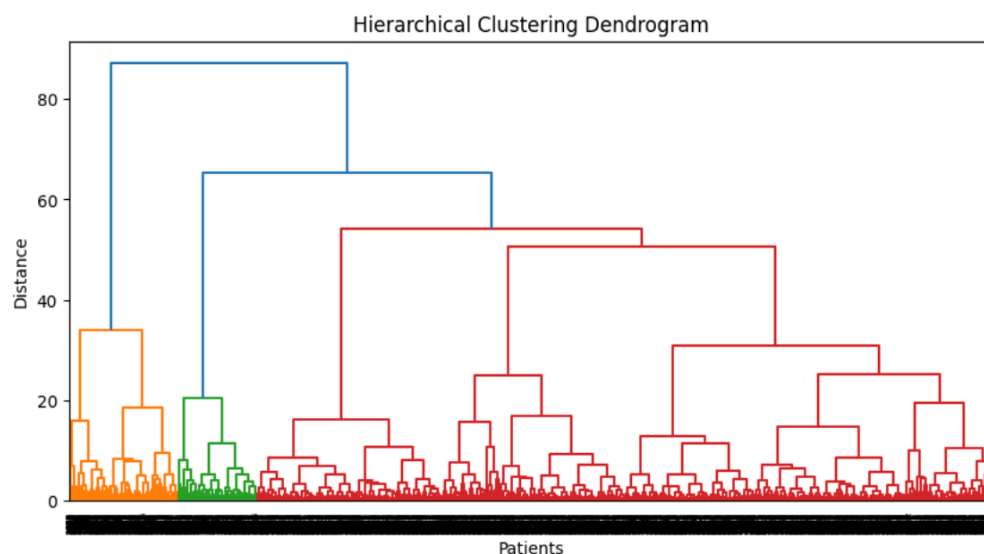


Figure 5: Patient clusters based on key health indicators

The dendrogram above created four clusters of orange, green, red, and ark orange. The y-axis shows distance which represents how similar or dissimilar patients are. The short branches which merge and low height represent high similarity, those who merge at a higher height represent lower similarity. Along with the dendrogram we got four clusters with varying stroke rates. Cluster 1 has the highest stroke rate at 14%, then cluster 2 a 12.6%. Following the first two clusters there is a drop off where cluster 4 has a stroke rate of 4.2% and then cluster 3 with a stroke rate of 0.15%. This means that clusters 1 and 2 have health profiles that put them at a higher risk which can indicate those patients having high BMI, high glucose level, and of older age. Cluster 3 represents patients who are low risk which can indicate younger patients with normal glucose levels and a healthy BMI. The clustering allows for identify high risk patient groups to be able to intervene and put preventative measures in to reduce stroke risk. Having identified patient clusters and stroke risk patterns, I can now examine one of the most important variables being the average glucose level.

I specifically examined whether glucose levels differ between patients who experienced a stroke and those who didn't, across different professions. By grouping patients based on their stroke status and profession, i can explore how occupational factors may affect glucose levels and stroke risk. I completed this analysis of this question in RStudio as it is easier to compute statistics and create graphs. To showcase and answer this

question I found the mean glucose level for each profession. Next, I created a box and whisker plot which compared the average glucose level across the different professions analyzing those who have had strokes and those who haven't in each profession.

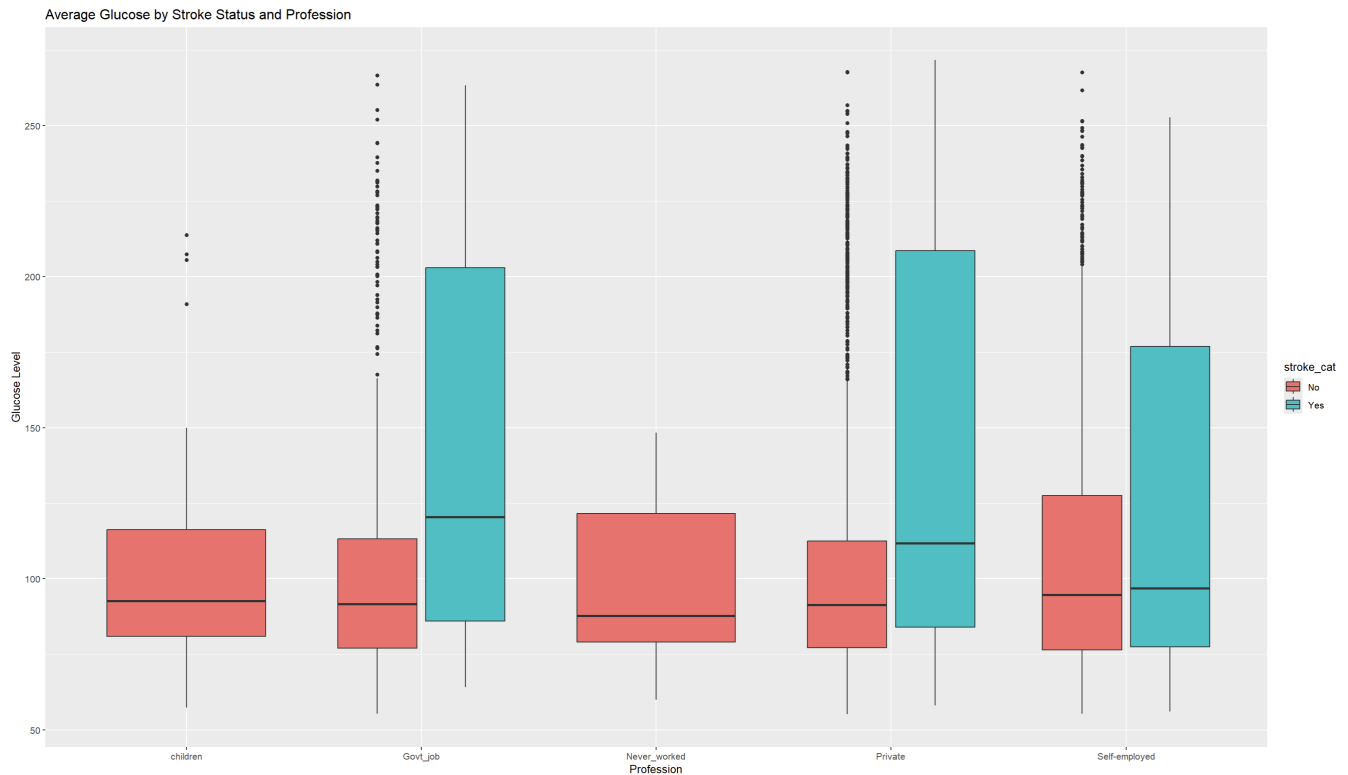


Figure 6: Average Glucose Level by Stroke and Profession

Observing the graph we can see that patients who have had a stroke have higher average glucose levels compared to those who have not had a stroke. This difference is greatly shown between two professions in the government and private sector. Stroke patients had significantly higher medians and variability in the average glucose levels compared to patients with no stroke in those professions. We can also see that the average glucose level for patients who have not had strokes is very similar across the different professions. We can also see many outliers for patients who have not had strokes and this can indicate that they are potentially at a high risk of having a stroke because they have a high glucose level and that is a important factor when looking at risk of a stroke. Overall, these findings highlight patterns in stroke risk and patient profiles.

The analysis shows that BMI, age, and average glucose level are among the most important factors when talking about stroke risk in patients. Hierarchical clustering helped identify different profiles which showed that patients can be grouped based on similar characteristics that influence stroke risk. Then the comparison of average glucose level across professions and stroke status suggested that occupation does not have an affect on glucose level and that the other factors play a much more important role. All together these

findings provide a deeper understanding of factors that affect stroke risk and why its important to target certain patient profiles to take preventative measures. It also showed how data science and machine learning can be used in the medical field to help find these groups of people and outliers. With the overall goal to implement prevention plans to reduce stroke occurrence across populations.

Sources

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